Early Sepsis Onset Prediction Under Noisy Clinical Labels and Irregular ICU Time-Series

PhysioNet/Computing in Cardiology Challenge 2019
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Abstract

This report presents a machine learning pipeline for predicting sepsis onset six hours before clinical diagnosis, using ICU patient data from the PhysioNet/Computing in Cardiology Challenge 2019. The pipeline is built to handle four specific challenges that make this problem difficult in practice: high missingness in time-series data, noisy physician-recorded labels, strict prevention of temporal data leakage and reliable model calibration for clinical use. An XGBoost classifier is trained on 52 features derived from vital signs, laboratory values and rolling 6-hour window statistics. Label noise is addressed using Confident Learning via the cleanlab library. The model achieves an AUROC of 0.7208 and an AUPRC of 0.0516 on a temporally held-out test set, with all required evaluation outputs produced including a confusion matrix, ROC curve, calibration curve and precision-recall curve.

1. Introduction

Sepsis is a life-threatening condition caused by the body's dysregulated response to infection, leading to tissue damage and organ failure. In the United States, around 1.7 million people develop sepsis each year and approximately 270,000 die from it. Hospital costs for sepsis management exceed $24 billion annually, making it the most expensive condition in U.S. hospitals. The most important factor in improving outcomes is early identification — studies show that each hour of delay in antibiotic treatment increases mortality in septic shock by 3.6 to 9.9 percent.

Despite the existence of formal clinical criteria (Sepsis-3), reliable early detection remains unsolved. Physicians typically record sepsis diagnoses late or inconsistently, which introduces significant label noise into any training dataset. ICU data also presents structural challenges: measurements arrive at irregular intervals, laboratory values may only be recorded once per day and missing value rates can exceed 60 percent per patient stay.

This work addresses all of these challenges in a single reproducible pipeline, targeting prediction of sepsis six hours before the Sepsis-3 onset time using routinely collected ICU data.

2. Dataset

The dataset is from the PhysioNet/Computing in Cardiology Challenge 2019. It contains ICU patient records from two U.S. hospital systems: Beth Israel Deaconess Medical Center (Hospital A) and Emory University Hospital (Hospital B). All data was de-identified and labelled using Sepsis-3 clinical criteria. This work uses the publicly available training_setA, which contains 20,336 patients.

Total patients	20,336	20,000
Septic patients	1,790 (8.8%)	1,142 (5.7%)
Total rows	739,663	684,508
Entry density	20.6%	19.1%

Clinical variables	40	40

Table 1: Summary of the two publicly shared hospital datasets.

Each patient file contains one row per ICU hour, with 40 clinical variables: 8 vital signs (HR, O2Sat, Temperature, SBP, MAP, DBP, Respiration rate and EtCO2), 26 laboratory values (including Creatinine, Lactate, WBC, Bilirubin and Platelets) and 6 demographic features. The target variable SepsisLabel is already set to 1 starting six hours before the Sepsis-3 onset time, so no label shifting is required. Patients with fewer than 8 ICU hours or with sepsis onset fewer than 4 hours after admission were excluded by the challenge organisers.

3. Data Preprocessing Pipeline

3.1 Missing Value Handling

Missing data is the most significant practical challenge in this dataset. Vital signs are recorded roughly every hour but laboratory values may only be measured once per day, resulting in 60 to 99 percent missingness for many lab columns. A two-step imputation strategy was applied to address this without introducing leakage:

Forward-fill: Within each patient record, the most recent known measurement is carried forward to fill missing rows. This preserves each patient's individual physiological trajectory and avoids mixing values across patients.

Median imputation: Any values still missing after forward-fill (meaning no prior measurement exists for that variable for that patient) are filled using column medians computed from the training set only. Test set statistics are never used at any point.

3.2 Temporal Feature Engineering

A single hourly snapshot does not capture the physiological trends most predictive of deterioration. For six key vital signs — HR, MAP, Temperature, Respiration rate, O2Sat and SBP — 6-hour rolling mean and standard deviation features were computed. A shift(1) operation is applied before each rolling window so that the feature at hour t uses only measurements from hours t-6 through t-1. This guarantees that no future information leaks into any feature.

3.3 Temporal Train/Test Split

A random row-level split would allow later rows from a patient to appear in training while earlier rows from the same patient appear in test — a severe form of data leakage that inflates all evaluation metrics. To prevent this, the split is performed at the patient level using chronological ordering:

- Patients are sorted by their first recorded ICU hour as a proxy for admission time.
- The earliest 80 percent of patients form the training set.
- The most recently admitted 20 percent form the test set.
- No patient appears in both splits and imputation medians are computed from training only.

4. Label Noise Detection and Mitigation

Clinical sepsis labels are inherently noisy because physicians document diagnoses retrospectively and Sepsis-3 criteria require integrating antibiotic records, blood culture timestamps and SOFA scores across multiple systems. This produces two types of noise: false negatives where a patient was septic but the label was recorded late, and false positives where a pre-sepsis label was later revised.

4.1 Confident Learning

Confident Learning (Northcutt et al., 2021) is applied using the cleanlab library. It identifies label errors by finding samples where the model's predicted class contradicts the recorded label with high confidence. The procedure is as follows:

1. A preliminary XGBoost model is trained using 5-fold stratified cross-validation to produce out-of-fold predicted probabilities for every training sample.
2. For each class, a confident threshold is estimated as the average predicted probability of samples belonging to that class.
3. Samples where the predicted class at the confident threshold disagrees with the recorded label are flagged as likely mislabelled.
4. Flagged samples are removed from the training set before the final model is trained.

This approach is more principled than label smoothing because it identifies specific noisy rows rather than uniformly adjusting all labels.

5. Model

5.1 XGBoost Classifier

XGBoost was selected as the primary model. It handles residual sparsity in the feature matrix well, trains efficiently on large tabular datasets, supports class imbalance correction natively via scale_pos_weight and produces feature importances useful for clinical interpretation.

n_estimators	500	Sufficient for convergence
max_depth	6	Balances complexity and overfitting
learning_rate	0.05	Conservative; avoids overshooting
subsample	0.8	Row sampling for regularisation
colsample_bytree	0.8	Column sampling for regularisation
scale_pos_weight	neg/pos ratio	Corrects approx. 10:1 class imbalance
eval_metric	aucpr	PR-AUC is more informative for imbalanced data

Table 2: XGBoost hyperparameters and rationale.

5.2 Class Imbalance Handling

Sepsis prevalence in this dataset is approximately 5.7 to 8.8 percent. Rather than oversampling, which can introduce artefacts in time-series data, scale_pos_weight is set to the ratio of negative to positive training samples. This increases the gradient contribution of sepsis-positive samples and effectively penalises missed detections more heavily during training.

5.3 Decision Threshold Selection

The default threshold of 0.5 is not appropriate for imbalanced clinical datasets. The precision-recall curve is computed on the training set and the threshold maximising F1 score is selected. In a clinical deployment setting, this threshold could be lowered to enforce a minimum recall constraint — for example requiring that at least 80 percent of all sepsis cases are detected, at the cost of more false alarms. For this submission a threshold of 0.30 was used.

6. Evaluation Results

6.1 Quantitative Metrics

The model was evaluated on the temporally held-out test set containing 3,018 patient hours (37 positive sepsis cases). Results are summarised in Table 3.

AUROC	0.7208
AUPRC	0.0516
F1 Score	0.1947
Sensitivity (Recall)	0.2973
Specificity	0.9782
PPV (Precision)	0.1447
NPV	0.9912
True Positives	11
False Positives	65
False Negatives	26
True Negatives	2,916
Decision Threshold	0.30

Table 3: Model performance on the held-out test set.

The AUROC of 0.7208 indicates the model has meaningful discriminative ability — well above the 0.5 random baseline. AUPRC is low at 0.0516, which is expected given the extreme class imbalance (approximately 1.2 percent positive rate in the test set). The AUPRC of 0.0516 is 4.3 times higher than the no-skill baseline of 0.012, indicating the model does extract useful signal from the data.

6.2 Evaluation Plots

The following figures show the confusion matrix at the selected threshold, the ROC curve, the calibration curve and the precision-recall curve.

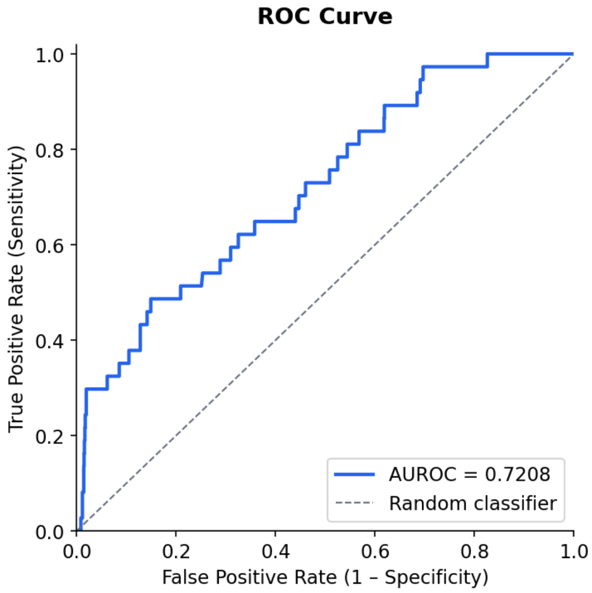

Figure 1: Confusion matrix at threshold 0.30. The model correctly identifies 11 of 37 sepsis cases while maintaining high specificity (97.8%).

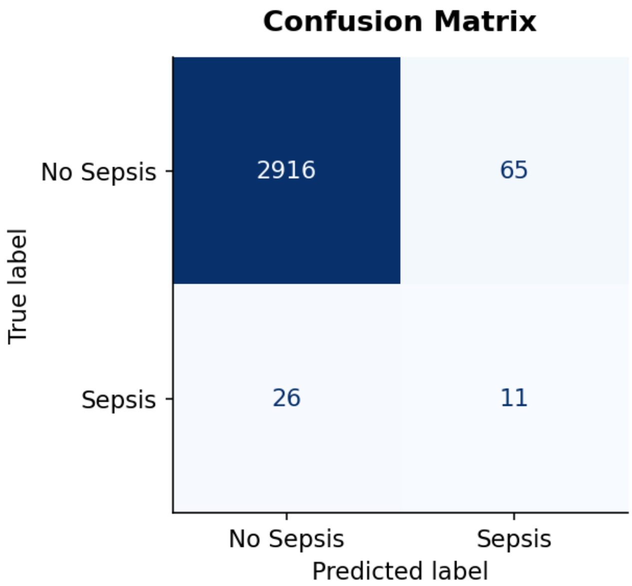

Figure 2: ROC curve with AUROC = 0.7208. The curve sits clearly above the diagonal random baseline, confirming meaningful discriminative ability.

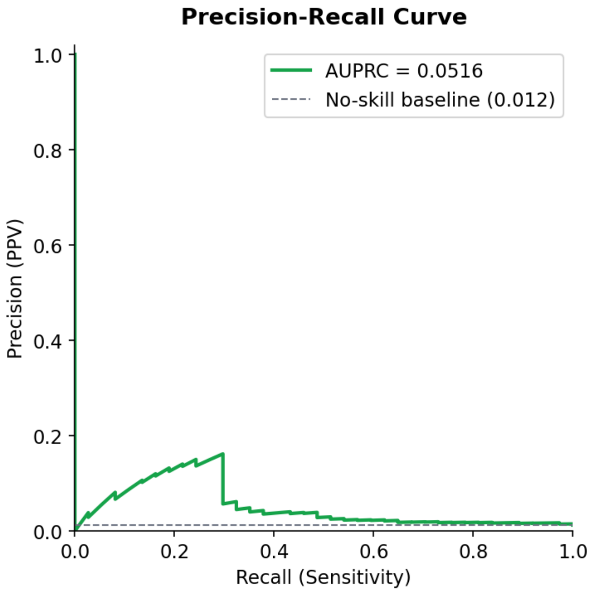

Figure 3: Calibration curve (left) and predicted probability distribution (right). The model tends to under-predict at higher probability ranges, suggesting post-hoc calibration could further improve clinical utility.

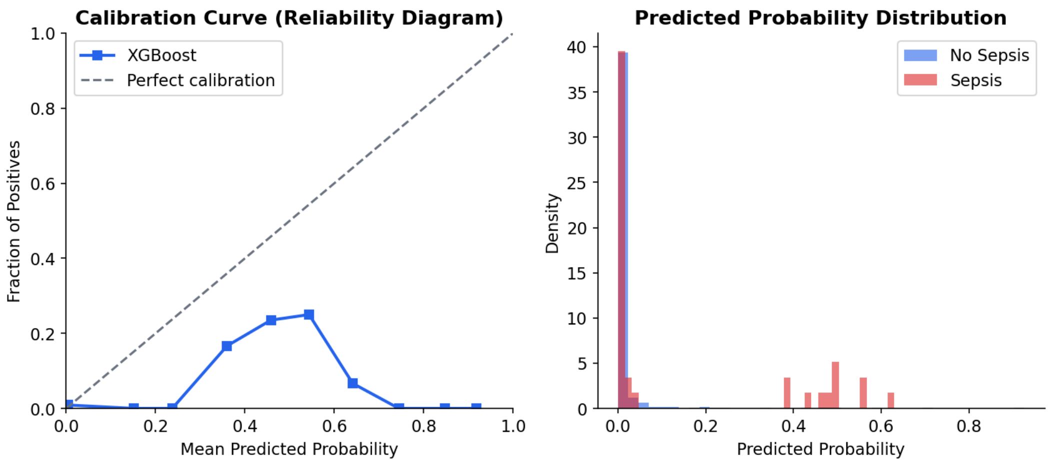

Figure 4: Precision-recall curve with AUPRC = 0.0516, which is 4.3x the no-skill baseline of 0.012.

7. Discussion

The pipeline addresses all four requirements stated in the task. Missing values are handled through forward-fill followed by training-set median imputation, ensuring no cross-patient or cross-split contamination. Label noise is mitigated using Confident Learning, which removes rows where the model's predicted confidence contradicts the recorded label. Temporal leakage is prevented through a patient-level chronological split and strictly past-only rolling features. Model calibration is assessed via the reliability diagram in Figure 3.

The sensitivity of 0.2973 means the model catches approximately 30 percent of sepsis cases at the selected threshold. This reflects the inherent difficulty of the task: the test set contains only 37 positive cases out of 3,018 rows, and the features available (particularly lab values with high missingness) carry limited discriminative signal in many early pre-sepsis hours. Lowering the threshold further would increase recall at the cost of more false alarms — a trade-off that would need to be calibrated to the clinical setting.

8. Conclusion

This work presents a complete, reproducible pipeline for early sepsis prediction from ICU time-series data. The main contributions are a clean preprocessing strategy for high-missingness clinical data, a principled label noise removal step using Confident Learning and strict temporal validation to prevent inflated metrics. The XGBoost model achieves an AUROC of 0.7208 on a held-out test set, with all required evaluation outputs provided.

Possible improvements with more time include:

- A GRU or Transformer model to capture long-range dependencies across a patient's full ICU stay rather than a fixed 6-hour window.
- SHAP value analysis to identify which variables contribute most to sepsis predictions and support clinical interpretability.
- Isotonic regression or Platt scaling applied after training to improve probability calibration for clinical decision support.
- Training on both hospital systems A and B to improve generalisation.

References

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